Trailing Labor Yield: A White Paper

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Abstract

Most contributor upside is trapped in illiquid equity or paid in volatile tokens that workers must sell. Trailing Labor Yield (TLY) is a stablecoin- denominated compensation mechanism that pays normal cash compensation while active, adds a small bonus tied to cumulative historical pay, and converts the final active bonus into a fixed-duration trailing payout when the contributor exits. The payout tapers over time, giving former contributors continued economic exposure to organizational survival without issuing governance tokens or creating untapered pension-like claims. TLY is not appropriate for every organization; it requires reserve planning, transparent compensation governance, and jurisdiction-specific legal review.

Executive Summary

- TLY offers an alternative to the usual tradeoff between illiquid equity and volatile token compensation.
- Contributors receive base pay while active, plus an active bonus equal to a small share of their historical pay pool.
- When a contributor leaves, their most recent active bonus is snapshotted into a trailing stablecoin payout.
- The trailing payout tapers, for example by 5 percent annually, and expires after a fixed period, for example 25 years.
- The taper makes the alumni liability a runoff process rather than a perpetuity.
- The EVM implementation uses per-wallet state and pull claims, so the treasury never loops over all alumni.
- Active contributor pay can be prioritized during stress by pausing trailing claims.
- TLY is a mechanism design proposal, not legal, tax, accounting, or investment advice.

Who This Is For

This paper is written for founders, DAO operators, contributor pods, protocol treasurers, compensation committees, and workers negotiating long-term upside. It assumes basic familiarity with stablecoins, treasury management, and smart contract-based payroll, but it does not require reading Solidity or the Python simulator before understanding the mechanism.

Plain-English Mechanism

TLY pays contributors normal cash compensation while they are active. It also grants a small bonus tied to cumulative historical pay. When a contributor leaves, their final active bonus is converted into a trailing stablecoin payout for a fixed number of periods. That payout tapers over time, so former contributors retain upside without creating permanent, growing liabilities for the organization.

The mechanism has two states:

1. Active state: the contributor receives base pay and earns a compounding active bonus.
2. Legacy state: the contributor no longer earns base pay, but can claim a trailing payout based on their final active bonus.

The key design choice is that TLY pays in a treasury reserve asset, such as a stablecoin, rather than a governance token. It separates labor compensation from governance-asset dilution.

Mechanism At A Glance

TLY can be stated in three steps:

1. While active, the contributor receives base pay and earns an active bonus.
2. On exit, the protocol snapshots the contributor's final active bonus as the initial trailing amount.
3. In legacy state, that amount is paid out through a tapering claim schedule over a fixed duration.

A Worked Example

Consider one contributor paid annually.

Input	Value
Base pay	$100,000 per year
Historical pay pool entering year	$250,000
Active accrual share	1%
Annual taper after exit	5%
Trailing duration	25 years

The active bonus for the year is:

$$\text{Active Bonus} = (\$250,000 + \$100,000) \times 0.01 = \$3,500\,.$$

If the contributor exits after that year, the protocol snapshots the final active bonus as the initial trailing amount:

$$R = \$3,500\,.$$

With a 5 percent annual taper, the first three trailing payouts are:

Year after exit	Payout
1	$3,325.00
2	$3,158.75
3	$3,000.81

The total undiscounted payout over 25 years is approximately:

$$\sum_{s=1}^{25} 3500 \times 0.95^{s} \approx \$48,054\,.$$

Bounded means the obligation runs off structurally rather than compounding forever; it does not mean every parameter set is prudent or fully prefunded. This is not a promise that every organization can afford this schedule. It is a transparent rule that can be simulated before adoption. At the organization level, the burden depends on payroll growth, turnover, active accrual rate, trailing duration, and the taper.

The Compensation Problem

Traditional startup equity solves cash preservation for the company but often fails workers. The worker receives paper exposure to a private capital structure with uncertain liquidity, exercise costs, preferred-share overhang, and limited control. The claim is called ownership, but for many workers it behaves like a long-dated option with unclear realizable value.

DAO token compensation improves liquidity but introduces reflexivity. If contributors receive governance tokens and sell them for living expenses, the protocol pays labor by pressuring the same asset that coordinates governance. If the token price falls, more tokens are required to pay the same real wage. That can create a procyclical compensation loop.

Pensions and perpetuities fail in the opposite direction. They reduce worker uncertainty by creating employer uncertainty. Fixed, permanent obligations can accumulate into a large inactive claimant base. This is the Iceberg Effect: the visible current payroll is only the surface; the hidden mass is the stock of future obligations.

TLY sits between these structures. It gives workers a portable claim on future treasury cash flows, but the claim tapers and expires.

What TLY Is

TLY is a compensation rule that converts labor history into a trailing cash-flow claim.

- It is not equity because it does not transfer governance or ownership.
- It is not ordinary token compensation because it does not require issuing or selling the governance asset.
- It is not a traditional pension because the payout is based on the final active bonus and tapers over a fixed horizon.
- It is not a generic revenue share because the claim is contributor-specific and derives from historical labor compensation.

TLY provides survival-contingent cash-flow exposure, not the uncapped capital-appreciation upside of equity.

In the reference design, the active bonus is:

$$\text{Active Bonus} = (\text{Historical Pay Pool} + \text{Base Pay}) \times \text{Accrual Share}\,.$$

When the contributor exits, the final active bonus becomes the initial trailing amount. The trailing claim then pays:

$$\text{Trailing Payout}_s = \text{Initial Trailing Amount} \times \rho^s,$$

where s is the number of epochs after departure and ρ is the per-epoch taper factor. For annual epochs and a 5 percent taper, $\rho = 0.95$.

Why The Liability Stays Bounded

The liability stays bounded because old claims lose weight over time. A trailing claim with initial amount R, taper factor $\rho < 1$, and duration K has total undiscounted payout:

$$\sum_{s=1}^{K} R\rho^s = R\rho\frac{1-\rho^K}{1-\rho}\,.$$

At a 5 percent annual taper and 25-year duration:

$$\rho = 0.95,$$

and:

$$R \times 0.95 \times \frac{1-0.95^{25}}{0.05} \approx 13.73R\,.$$

This number is the lifetime multiple of the final active bonus, not of base salary. The final active bonus is itself controlled by the active accrual share, typically modeled at 1 percent.

At the organization level, three forces interact:

1. Low active accrual limits new trailing claims.
2. Payroll growth makes old cohorts smaller relative to current payroll.
3. The taper reduces the weight of every legacy cohort over time.

Under the simulator's baseline growth intuition of roughly 4 percent gross payroll growth and the EVM convention that the first legacy claim receives $R\rho$, the approximate steady burden is:

Turnover	Active bonus ratio	Approx. trailing burden
8%	9.35%	7.9% of payroll
10%	7.92%	8.3% of payroll
30%	3.12%	9.8% of payroll

These values are illustrative rather than universal. The important point is structural: the taper converts a potentially permanent alumni obligation into a stationary runoff process. No artificial cap on an individual contributor's historical pay pool is needed for stationarity, although organizations may still adopt governance controls around pay changes and exit timing.

Design Conditions

TLY is most defensible when the following conditions hold:

- low active accrual share;
- taper factor below one;
- finite trailing duration;
- credible treasury funding policy;
- governance controls around exit timing and pay changes.

Comparison With Adjacent Mechanisms

Feature	Startup equity	DAO token comp	Pension / perpetuity	Revenue share	TLY
Liquid while active	Usually no	Usually yes	No	Sometimes	Yes, for base and active pay
Direct governance dilution	Often yes	Yes	No	No	No

Feature	Startup equity	DAO token comp	Pension / perpetuity	Revenue share	TLY
Treasury cash obligation	No near-term cash	Often indirect	Yes	Yes	Yes
Bounded long-run liability	Cap table bounded, value uncertain	Issuance may expand	Often weak or no	Depends on contract	Yes, via taper and term
Portable after exit	Sometimes, but illiquid	Yes	Yes	Sometimes	Yes
Depends on market price	High	High	Low	Medium	Low if paid in stablecoins
Depends on treasury solvency	Indirect	Indirect	High	High	High
Legal / tax complexity	High	High	High	Medium to high	Medium to high
Fit for pre-revenue startups	Strong	Common but volatile	Weak	Weak	Limited unless prefunded, delayed, or threshold-activated

TLY is not categorically better than these mechanisms. It is designed for a narrow target: liquid labor upside paid outside the governance asset, with long-run obligations tapered by rule.

Where TLY Fits Best

TLY is most attractive for organizations that already have, or can reasonably develop, stable treasury capacity.

Good candidates include:

- mature DAOs with recurring protocol revenue;
- profitable crypto-native firms;
- protocol-adjacent service organizations;
- cooperatives or mission-driven organizations with reserve discipline;
- treasury-backed ecosystems that want alumni alignment without token sell pressure.

Poor candidates include:

- pre-revenue startups seeking immediate live-payout TLY, unless prefunded, delayed, or threshold-activated;
- organizations with highly volatile treasuries and no stable reserve policy;
- teams unwilling to formalize compensation governance;
- firms that need pure upside asymmetry rather than bounded cash obligations;
- organizations that cannot manage jurisdiction-specific labor, tax, and deferred-compensation requirements.

Operational Design And Governance Rules

The reference EVM architecture has two contracts:

Contract	Role
`ContributorRegistry.sol`	stores contributor status, base pay, historical pay pool, final active bonus, departure epoch, and claim checkpoints
`TreasuryDistributor.sol`	holds the payment token, computes active compensation, computes trailing claims, and transfers funds

The system uses per-contributor accounting. The registry stores each contributor's `historicalPayPool`, `lastActiveBonus`, `initialLegacyAmount`, `departureEpoch`, and `lastLegacyClaimEpoch`.

The treasury uses pull payments:

- active contributors call `claimActiveComp()`;
- legacy contributors call `claimTrailingYield()`.

The treasury never iterates through all alumni. That is essential for gas safety.

The governance design separates roles:

- `POD_ADMIN_ROLE`: operational sub-DAO or multisig that can add contributors, remove contributors, and update base pay.
- `DAO_ADMIN_ROLE`: treasury-level role that can pause and unpause legacy claims during stress.

If legacy claims are paused, active compensation remains claimable. This keeps current labor funded during bear markets or treasury stress.

Capitalization Models For TLY

The pre-revenue question is central. TLY is a cash-flow claim. It should not be adopted casually by organizations that lack a path to funding future claims.

Possible capitalization models include:

1. Reserve-funded model: the treasury prefunds expected trailing claims from stable reserves.
2. Margin-funded model: each epoch's claims are paid from operating surplus.
3. Hybrid model: early-stage contributors receive a small TLY schedule, with larger accrual activated after revenue maturity.
4. Milestone activation: legacy payouts begin only after revenue, treasury, or runway thresholds are met.
5. Buffer-pool model: a portion of active payroll or operating margin funds a reserve pool for future trailing claims.

These models can be combined. For example, an early protocol might accrue TLY internally but delay claim activation until a stablecoin reserve threshold is met. The design question is not whether the claim can be written on-chain. It is whether the organization has a credible funding policy.

Risks, Limitations, And Failure Modes

Treasury Solvency And The Cash-Flow Paradox

TLY is not for cash-starved organizations with no realistic treasury path. It works best for profitable, treasury-backed, or reserve-capitalized entities. Pre-revenue organizations may need delayed activation, reserve escrows, smaller initial accrual rates, or hybrid token-cash structures.

Trailing claims should be described precisely. Depending on legal structure, they may be contingent claims rather than guaranteed debt. If treasury funds are insufficient, the organization must define whether missed claims accrue, queue, partially defer, or remain claimable only when the treasury is funded.

Legal, Tax, And Deferred Compensation Risk

TLY may trigger deferred compensation, payroll, securities, labor, tax, employment-benefit, or accounting issues depending on jurisdiction. In some jurisdictions, the legacy claim may need to be implemented through an off-chain employment, contractor, deferred-compensation, or bonus-rights agreement rather than treated as a purely on-chain entitlement. Actual deployment may require wrappers, employer-of-record structures, written off-chain agreements, entity design, and tax withholding processes.

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Terminal Bonus Manipulation

If the trailing claim is based only on the last active bonus, parties may try to manipulate the terminal period. Defenses include:

- using a rolling average of recent active bonuses instead of a single final bonus;
- locking base pay changes near expected departure;
- requiring notice periods before legacy conversion;
- requiring governance review for exceptional compensation changes;
- capping snapshot growth relative to a trailing average, while still avoiding a cap on the contributor's historical pay pool itself.

The current reference contract snapshots `lastActiveBonus`. Production deployments may choose a more conservative snapshot policy.

Treasury Stress And Insolvency

The Solidity reference includes `pauseLegacyClaims()` so a DAO admin can block legacy withdrawals while preserving active compensation. This is a priority rule, not a complete insolvency framework.

Before deployment, organizations should define:

- whether paused claims accrue or simply wait;
- whether claimants can later catch up using aggregate claims;
- whether partial payment is allowed;
- whether legacy claims are legally enforceable debt or contingent protocol benefits;
- what disclosures contributors receive before accepting the structure.

Governance Abuse And Favoritism

TLY depends on credible compensation administration. A pod admin that can update base pay or time departures can influence future trailing claims.

Mitigations include:

- transparent compensation policies;
- multisig-controlled contributor administration;
- published parameter changes;
- auditable events for base pay updates and exits;
- notice periods and review windows near departure;
- periodic reporting of active payroll, active bonuses, and projected trailing burden.

Parameter Risk

The mechanism is bounded only under coherent parameters. A high accrual rate, weak taper, very long duration, or low-growth payroll can create excessive cash obligations. The simulator should be used before governance adopts or changes parameters.

Implementation Path And Pilot Model

A practical pilot can be small and explicit:

1. Choose the payment asset, usually a stablecoin or treasury reserve asset.
2. Choose epoch length, such as monthly or annual.
3. Choose an initial accrual share, such as 1 percent.
4. Choose a trailing duration, such as 25 annual epochs or 300 monthly epochs.
5. Convert the annual taper into a per-epoch WAD factor.
6. Simulate expected burden under growth, wage, turnover, revenue, and margin assumptions.
7. Define treasury funding policy and stress rules.
8. Define compensation governance, including pod admin authority and review rules near departure.
9. Deploy contracts or begin with an off-chain legal agreement that mirrors the same equations.
10. Publish recurring reports comparing actual burden against simulated burden.

The repository contains two practical tools:

- `sim/app.py`: Streamlit dashboard for scenario analysis.
- `dao/contracts/`: EVM-compatible reference contracts for registry and treasury distribution.

Legal And Regulatory Disclaimer

This document is for mechanism design and technical research only. It is not legal, tax, accounting, employment, securities, benefits, or investment advice. On-chain enforceability does not eliminate off-chain legal obligations. Organizations considering TLY should obtain jurisdiction-specific advice before deployment.

Conclusion

TLY offers an alternative to the common compensation tradeoff between illiquid equity and volatile token issuance. It can separate contributor upside from governance-asset dilution while giving the organization a modeled, tapering obligation.

The active state rewards tenure through compounding accrual. The legacy state turns the final active bonus into a trailing stablecoin claim. The taper turns an alumni obligation into a runoff process. The

EVM design maps this process to per-wallet state, pull-based claims, bounded missed-epoch processing, and stress controls for legacy withdrawals.

The claim is not free. It must be funded, governed, disclosed, and legally structured. But for reserve-disciplined organizations, TLY can create a more legible bargain: contributors receive liquid, trailing upside; the organization preserves governance integrity and can model the long-run burden before making the promise.

Appendix A: Full Formulas

Lowercase subscripts refer to contributor-level quantities. Aggregate quantities omit contributor indexing. Unless otherwise stated, all quantities are per epoch.

A.1 Notation

- $B_{i,t}$: contributor i's base pay in epoch t.
- $H_{i,t}$: contributor i's historical pay pool entering epoch t.
- α: active accrual share.
- $A_{i,t}$: active bonus.
- R_i: initial trailing amount after exit.
- δ: taper rate.
- $\rho = 1 - \delta$: per-epoch taper factor.
- K: maximum trailing duration.
- q: turnover rate.
- G: gross payroll growth factor.
- L_t: aggregate trailing payout in epoch t.

A.2 Active State

Contributor-level active bonus:

$$A_{i,t} = \alpha(H_{i,t} + B_{i,t}) .$$

Contributor-level historical-pool update:

$$H_{i,t+1} = H_{i,t} + B_{i,t} + A_{i,t} .$$

Aggregate simulator update with turnover:

$$H_{t+1} = (H_t + B_t + A_t)(1 - q) .$$

A.3 Legacy State

Exit snapshot:

$$R_i = A_{i,\tau_i}.$$

Trailing payout s epochs after departure:

$$Y_{i,s} = R_i \rho^s.$$

Total finite-horizon trailing payout:

$$\sum_{s=1}^{K} R_i \rho^s = R_i \rho \frac{1-\rho^K}{1-\rho}.$$

Infinite-horizon upper bound:

$$\sum_{s=1}^{\infty} R_i \rho^s = R_i \frac{\rho}{1-\rho}.$$

A.4 Aggregate Burden Approximation

New legacy cohort from turnover:

$$N_t = qA_t.$$

Aggregate trailing payout:

$$L_t = \sum_{j=1}^{K} N_{t-j} \rho^{j-1}.$$

Payroll growth:

$$B_{t+1} = GB_t.$$

Active bonus ratio:

$$m_t = \frac{A_t}{B_t}.$$

Historical-pool ratio:

$$h = \frac{H_t}{B_t}.$$

Steady-state relationship:

$$h = \frac{(1-q)(1+\alpha)}{G}(h+1).$$

Let:

$$c = \frac{(1-q)(1+\alpha)}{G}.$$

If $c < 1$:

$$h = \frac{c}{1-c}$$

and:

$$m = \frac{\alpha}{1-c}.$$

Approximate burden ratio:

$$\frac{L_t}{B_t} \approx qm \sum_{j=1}^{K} \frac{\rho^{j-1}}{G^j}.$$

Infinite-horizon upper bound:

$$\frac{L_t}{B_t} \leq \frac{qm}{G-\rho}.$$

If the first post-departure claim receives $R_i\rho$, as in the EVM convention:

$$\frac{L_t}{B_t} \leq \frac{qm\rho}{G-\rho}.$$

Appendix B: Simulation Assumptions

The Python simulator in `sim/app.py` and `sim/baseline_dollars.py` models the macro version of TLY using cohorts.

Core inputs include:

- `horizon_periods`;
- `initial_workers`;
- `worker_growth`;
- `wage_per_worker`;
- `wage_growth`;
- `accrual_share`;
- `turnover_rate`;
- `legacy_payout_years`;
- `annuity_decay_rate`, which corresponds to the TLY taper rate;
- `revenue_multiplier`;
- `gross_margin`.

The simulator computes:

```
base_payroll = workers * wage
active_bonus = (historical_pay_pool + base_payroll) * accrual_share
legacy_payout = cohort_amount * (1 - taper_rate) ^ cohort_age
historical_pay_pool_next =
    (historical_pay_pool + base_payroll + active_bonus) * (1 - turnover_rate)
```

The simulator is intentionally stylized. It does not model legal enforceability, tax treatment, default litigation, stablecoin depegging, claimant mortality, or endogenous contributor behavior. It is a planning tool for burden dynamics, not a solvency guarantee.

Appendix C: Smart Contract Architecture

The reference EVM implementation lives in `dao/contracts`.

C.1 Contributor Registry

`ContributorRegistry.sol` stores:

- contributor status: none, active, legacy, removed;
- payout address;
- base pay per epoch;
- historical pay pool;
- last active bonus;
- initial legacy amount;
- joined epoch;
- last settled active epoch;
- departure epoch;
- last legacy claim epoch.

`POD_ADMIN_ROLE` can:

- add contributors;
- update payout addresses;
- update base pay;
- remove contributors into legacy status.

Only the configured treasury distributor can settle active epochs and mark legacy claims as paid.

C.2 Treasury Distributor

`TreasuryDistributor.sol` holds the payment token and computes claims.

Active claim:

```
activeBonus =
    (state.historicalPayPool +
state.basePayPerEpoch).mulWadDown(accrualShareWad);
```

Legacy claim:

```
uint256 taperMultiplierWad =
    WadMath.rpowWad(taperFactorPerEpochWad, elapsedEpochs);

claimable += state.initialLegacyAmount.mulWadDown(taperMultiplierWad);
```

The treasury uses OpenZeppelin `AccessControl`, `SafeERC20`, and `ReentrancyGuard`.

C.3 Pull Claims And Gas Safety

The system uses pull claims:

- `claimActiveComp()` for active contributors;
- `claimTrailingYield()` for legacy contributors.

Legacy missed-epoch claims are processed oldest-first and capped by `maxEpochsPerClaim`. This makes transaction cost independent of the total number of alumni.

C.4 Tranche Priority

`pauseLegacyClaims()` blocks legacy withdrawals while preserving active compensation. This supports active-pay priority during treasury stress.

Appendix D: FAQ And Objections

Is TLY just a pension?

No. A pension is typically a broad retirement benefit with employer-specific legal and actuarial treatment. TLY is a contributor-specific trailing cash-flow claim based on a final active bonus. It tapers and expires.

Is TLY just revenue share?

No. A generic revenue share usually pays a percentage of revenue. TLY pays a defined trailing amount based on contributor compensation history. It may be funded by revenue, reserves, margin, or a hybrid policy, but the claim formula is not a direct revenue percentage.

Why not just give equity?

Equity may be better for early-stage startups that need pure upside and cannot support cash obligations. TLY is better suited to organizations that want liquid, stablecoin-denominated labor upside without transferring governance or ownership.

Why not just use token vesting?

Token vesting is liquid and simple, but it ties compensation to token price and can create sell pressure on the governance asset. TLY can reduce that coupling by paying in a reserve asset.

What if the treasury fails?

TLY depends on treasury solvency. The reference implementation includes active-pay priority and pausable legacy claims, but this does not eliminate funding risk. Organizations must define whether claims are contingent, queued, partially paid, or legally enforceable.

Can someone game the exit bonus?

Yes, if governance is weak. Production deployments should consider rolling average snapshots, notice periods, review of exceptional pay changes, and rules that lock base pay near departure.

Does this create tax or legal issues?

Potentially. TLY may implicate deferred compensation, payroll, securities, labor, tax, benefits, or accounting rules. Legal design is jurisdiction- specific.

Why not cap historical pay?

The reference mechanism intentionally avoids artificial caps because long-tenure contributors are supposed to earn larger trailing claims. Aggregate stability comes from low active accrual, payroll growth, finite duration, and tapering legacy obligations. Governance may still restrict manipulation of terminal compensation.